

Lab setup for Mac users (Follow-up from Last Lecture)

For those using Apple products with Apple M1, M2, M3, M4, M5.

Key: Use UTM instead of VirtualBox

Required: brew, UTM

1. Install Kali Linux under UTM
 - a) Easiest way was from here (<https://mirror.bouwhuis.network/utm/>). I (hiraku-m) chose “Kali Linux 2025 UTM (Apple Silicon) (community image)” but another one should be also okay
2. Obtain Metasploitable 2 & 3 .vmdk files
 - a) Metasploitable 2 from NextCloud
 - Download Metasploitable.vmdk
 - b) Metasploitable 3 from <https://portal.cloud.hashicorp.com/vagrant/discover/rapid7/metasploitable3-win2k8>
 - Download 7.01GB file called “virtualbox” and add extension .zip, then unzip it. You get Metasploitable 3 in it.
3. Set Metasploitable on UTM (You do this twice for Metasploitable 2 and 3)
 - a) Convert .vmdk to .qcow2 on Terminal
 - brew install qemu
 - qemu-img convert -p -f vmdk -O qcow2 <filename>.vmdk <filename>.qcow2
4. Create VM in UTM (Refer Step 2 and 3 of https://medium.com/@d_mechraoui/how-to-run-ova-virtual-machine-on-apple-silicon-using-utm-m1-m2-m3-7a2a883697d4 You need to do a bit differently, though. See below)
 - a) Open UTM -> Create a New Virtual Machine -> Emulate -> Other -> Expert Mode [X86_64] [Standard PC (Q35 +...)] -> Boot Device [None]
 - b) See Step 4 of above link.

Connection between Kali Linux and Metasploitable should work fine in default. Check as follows:

1. Checking IP addresses for Kali Linux and Metasploitable2:
 - a. On Metasploitable2, run the command **ifconfig**. An example output is eth0 inet 192.168.64.x
 - b. On Kali Linux, run the command **ip addr**. An example output is eth0 inet 192.168.64.y
2. Connectivity test from Kali Linux to Metasploitable2:
 - a. Run the command **ping <Metasploitable2_IP>** on Kali Linux.
 - b. If you receive replies, it confirms that the network connection is working properly.